

Pranay Parepalli

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SUMMARY

I have expertise in Data Structures, Algorithms, and OOP, backed by 800+ solved problems on LeetCode and CodeChef. Hands-on experience building and refining machine learning models, from real-time data augmentation to dynamic learning rate scheduling, delivering robust and maintainable solutions in TensorFlow.

EDUCATION

Sardar Patel Institute of Technology (S.P.I.T)

Mumbai, India

B.Tech in Electronics and Telecommunication Engineering (7th Sem.)

Oct 2022 – Jul 2026

- Pursuing a Computer Science minor (Jan 2024 – Present)

S.V.P. Junior College of Science

Mumbai, India

Higher Secondary Certificate (HSC), 12th Grade

Oct 2020 – May 2022

SKILLS

Programming Languages: C, C++, Python

Machine Learning: TensorFlow, Keras, Scikit-learn, Deep Learning

Tools & Libraries: NumPy, Pandas, Matplotlib, Git, GitHub, Jupyter Notebook, Google Colab

Core Subjects: Data Structures and Algorithms (DSA), DBMS, OOPS, Computer Communication Networks (CCN)

Platforms: LeetCode, CodeChef, Kaggle

EXPERIENCE

Optificent Club, S.P.I.T

Aug 2024 – Present

Competitive Programming Mentor

Mumbai, India

- Mentored juniors in Competitive Programming and Code Optimization to boost contest performance.
- Conducted post-contest performance analysis to diagnose error patterns and define targeted improvement plans.
- Delivered personalized code reviews and one-on-one coaching to enhance problem-solving skills.

PROJECTS

AI X-Ray Diagnostics | *TensorFlow, Keras*

GitHub

- Developed a CNN model to classify chest X-ray images into pneumonia vs. normal categories.
- Applied real-time image augmentation to improve prediction accuracy on new data.
- Configured ReduceLROnPlateau to adjust learning rates based on validation loss.
- Achieved 98.2% accuracy and validated with confusion matrix and classification metrics.

Smart Food Image Classifier | *TensorFlow, ResNet50*

GitHub

- Fine-tuned ResNet50 to classify 101 food categories from the Food101 dataset.
- Implemented image augmentation strategies including flips, rotations, zoom, and brightness shifts.
- Trained model in two phases with early stopping to prevent overfitting.
- Achieved 75% validation accuracy with monitoring via loss curves.

Research-Forge | *Flask, TensorFlow, Scikit-learn, OpenAlex API*

GitHub

- Engineered a Flask web app for academic journal recommendation with session-based login.
- Implemented Sentence-BERT embedding and cosine similarity for semantic matching.
- Integrated OpenAlex API to retrieve relevant metadata and journal URLs.
- Enabled feedback-driven personalization by capturing user ratings and updating match scores.
- Enhanced recommendation relevance using rating-weighted logic.

ACHIEVEMENTS

- Solved over 800 algorithmic problems on LeetCode and CodeChef, improving logic and debugging skills.
- Ranked 324th out of 15,000+ participants worldwide in a CodeChef contest (Top 2.2%).
- Placed in the Top 10 at the university-level DataForge Hackathon.

HOBBIES

- Analyzing financial markets, economic trends, and long-term investing strategies.
- Solving algorithmic challenges and participating in coding competitions.